

Revocation Without Key Rotation

A single bonded token pair, minted and signed once; authority revoked mid-stream by flipping the mirror's state; the gate denies the next request without re-issuing the primary or touching any signing key.

Pair ID	bp-ce9581c1a64043ba
Primary token	AXIOM-BP-bp-ce9581c1a64043ba-A
Mirror token	AXIOM-BP-bp-ce9581c1a64043ba-B
Ledger path	/tmp/tmpnguqmojn/demo_ledger.jsonl
Generated at	2026-05-25T18:55:20.317539Z
Report version	0.1.0
Signing namespaces	axiom-bonded-pair-token-v1, axiom-bonded-pair-ledger-v1, axiom-report-v1

1. The minted pair

The two halves are signed once at mint time. Their content and signatures never change. State lives outside the signed bytes — in the ledger — so revocation is a register flip, not a re-issue.

Primary

```
{
  "pair_id": "bp-ce9581c1a64043ba",
  "partner_token_id": "AXIOM-BP-bp-ce9581c1a64043ba-B",
  "payload": {
    "execution_command": "run_local_model_optimization",
    "scope": "kid_toy_buddy_bear_v3"
  },
  "role": "primary",
  "signature": "423a0cbbd01f377d3b6cb56138c960fd33913088ba17a8856206b2238f83702c",
  "timestamp_ns": 1779735317853191328,
  "token_id": "AXIOM-BP-bp-ce9581c1a64043ba-A"
}
```

verifies: ✓ pair_verify(primary, mirror): ✓

Mirror

```
{
  "pair_id": "bp-ce9581c1a64043ba",
  "partner_token_id": "AXIOM-BP-bp-ce9581c1a64043ba-A",
  "payload": {
    "issued_to": "security_monitor",
    "monitor_target": "primary"
  },
  "role": "mirror",
}
```

```
{
  "signature": "9ccfec3bc5d256f0f98c2667ece3412a69b020aa292a5f02451a38565835a8a2",
  "timestamp_ns": 1779735317853191328,
  "token_id": "AXIOM-BP-bp-ce9581c1a64043ba-B"
}
```

2. State register — every transition

The ledger is append-only and hash-chained: every entry carries the previous entry's signature, so a flipped byte anywhere in the log breaks `verify_chain()` from that point forward.

from	to	actor	signature (truncated)
(init)	ACTIVE_VALIDATED	provisioner	567d55cec5a0da7d... 4f96d605809fdead e78ceeb70e7fb58a...
ACTIVE_VALIDATED	REVOKED	security_monitor	6e5f740282c9cea8

`verify_chain()`: **PASS** — every prev_signature link checks out.

3. Two gate checks of the SAME benign packet

A packet whose payload would normally pass the intent classifier (a benign request) is gated by the bonded-pair authority register. Between the two checks, only one thing changes: the ledger entry for this pair. The primary token's bytes and signature are identical in both checks.

Packet under test

```
{
  "destination": "model_runtime",
  "packet_id": "pkt-demo-001",
  "payload": {
    "pair_id": "bp-ce9581c1a64043ba",
    "text": "Tell me about how solar panels work."
  },
  "source": "buddy_bear_runtime"
}
```

Check 1 — pair is ACTIVE_VALIDATED

intent_class: INFORM · **blocked:** no · **confidence:** 0.55

signals: (none)

```
sig: 0b40ee3b7313af7a0b03676d628f8ae7e2f2427ae1f0c6c0a737f8a39314f00d
```

→ Operator runs **axiom-bonded-pair revoke bp-ce9581c1a64043ba**

The ledger appends one signed transition: ACTIVE_VALIDATED → REVOKED.

Check 2 — same packet, pair is REVOKED

intent_class: HARM · **blocked:** **yes** · **confidence:** 0.95

signals: bonded_pair_revoked, pair_id=bp-ce9581c1a64043ba, state=REVOKED

```
sig: 6b1df79890ab85226c0e0f4c7f9bc228e922c9ce23308f38500dfc1b23c5d212
```

The primary token's signature did not change. No key was rotated. No consumer was notified. The next gate request was denied because the gate consults the bonded-pair register, and the register has the new state.

4. What this is not

This primitive is not quantum entanglement. There are no Bell correlations and no non-local effects. The name in the implementation is “bonded paired-token with atomic state register.” The useful property is: authority can be revoked across an arbitrary distance with a single register-flip, without rotating the keys that signed the original tokens. That property is what the demonstration above is verifying byte-for-byte.

HMAC-SHA256 signed under axiom-report-v1. Verify with `scripts/verify_kid_audit.py --pdf <this.pdf> --sig <this.pdf.sig>` using the AXIOM_MASTER_KEY this file was generated with.